

Editors of Neural Networks

Dear Editors of Neural Networks,

Please consider our manuscript, "Quantization fragility under image corruption is recipe-dependent: paired evidence from INT8 and FP8 object detectors," for publication as a research article in Neural Networks.

Deployment of low-precision neural networks raises a reliability question that current evaluation practice answers ambiguously: whether an 8-bit quantized detector remains reliable under corrupted inputs depends on how the engine was built, not only on the datatype label. We make this concrete with a paired, codec-controlled protocol in which every corrupted input is paired with a JPEG-95 clean control and identical encoded bytes reach both precision treatments, and the corruption-associated change in the FP8-INT8 average-precision gap is estimated inside common-image bootstrap samples.

Three findings support the submission. First, under a frozen paired analysis of YOLO11n/m/x on KITTI, VOC, and a 2,000-image COCO subset, the interaction reverses sign across dataset-by-scale cells while FP8 exceeds INT8 on absolute AP in every block mean, so the corruption-induced change in the gap has no format-level ordering on this evidence. Second, a matched-recipe decomposition of RetinaNet localizes a severe INT8 collapse (-21.2 AP on KITTI) to regression-head quantization; excluding that head alone restores clean-competitive accuracy without retraining—the direction replicates on VOC, and both the RetinaNet localization and the contrasting FCOS boundary recur on off-the-shelf COCO-pretrained checkpoints that involve no evaluation-set checkpoint selection, while the 20-AP magnitude is a KITTI estimate that carries a disclosed development-contamination caveat. Third, a controlled two-fold calibration intervention—calibrating INT8 on corruption-augmented images and testing on held-out corruption families under complementary splits, with codec-matched calibration controls—shows the intervention itself is context-dependent (transferring in one fold-by-dataset cell, significantly harming another), so corruption-aware calibration is not supported as a reliable repair and clean-only calibration ranges are not supported as the dominant repairable cause. The paper therefore converts a measurement observation into a validated localization with a practical selective-precision recipe, whose deployment cost is bounded by an idle-gated single-host latency audit rather than left unevaluated.

The work fits Neural Networks' scope on the reliability and deployment behavior of neural network models: it treats quantization as a property of the executable recipe, identifies the fragile network region, and tests a calibrated intervention under frozen decision rules, rather than reporting a format leaderboard.

The manuscript is original, is not under consideration elsewhere, and all six authors have approved its submission. The research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Code and compact evidence accompany the submission as a hash-bound package deposited before submission as immutable release v3.0.2 (version DOI 10.5281/zenodo.23076132) under the all-versions concept DOI 10.5281/zenodo.22031663; raw datasets, checkpoints, built engines, and raw predictions are not redistributed, with their regeneration boundaries documented. The manuscript contains data-and-code, funding, competing-interest, CRediT, and generative-AI declarations.

Thank you for your consideration.

Sincerely,

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